Deep Learning Systems Analysis Report

RentShield Canada: NLP Detection of Risky Rental Listings

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Report Overview

This project focused on detecting renter-risk language in Toronto rental listings using deep learning. The dataset used was Toronto Kijiji Rental Data from Kaggle, which contains real rental advertisements with listing descriptions and housing details. A Transformer-based natural language processing model was implemented using DistilBERT for binary text classification. The goal was to classify listings into Safe or Risk categories based on rental listing language.

Dataset and Task Description

The dataset contains 4,106 rental listings and 23 columns. Important variables include Title, Price, Bedrooms, Bathrooms, Address, and Description. The most important text field for this project was Description because it contains the main rental advertisement language.

After removing rows with missing descriptions and duplicate text entries, 2,796 usable rows remained. A manually reviewed gold-standard labeled dataset was then created using sampled listings and targeted review examples. The final modeling dataset used 57 labeled rows for binary classification:

- Safe = 38
- Risk = 19

This dataset supports a supervised deep learning text classification task.

Dataset Source: Toronto Kijiji Rental Data (Kaggle)

https://www.kaggle.com/datasets/umeshkhatiwada/toronto-kijiji-rental-data

Model Architecture and Design Decisions

The baseline model used DistilBERT with a sequence classification head. DistilBERT is a compressed version of BERT that keeps strong language understanding performance while reducing model size and computational cost (Sanh et al., 2019).

DistilBERT was selected because:

- It is appropriate for text classification tasks
- It trains faster than full BERT
- It performs well on smaller hardware environments
- It is suitable for academic prototype work

The model output layer predicted two classes:

- Safe
- Risk

The AdamW optimizer was used with a learning rate of 2e-5. Fine-tuning pretrained Transformer models is a standard practice in NLP tasks (Devlin et al., 2019).

Experimental Comparison

One major aspect was changed in the experimental model: the loss function was modified using class weights.

Because the dataset had more Safe than Risk examples, weighted cross-entropy loss was used to give additional penalty when Risk samples were misclassified. This was selected to improve minority-class detection.

Baseline model:

- Standard loss

Experimental model:

- Weighted loss favoring Risk class

This was a meaningful comparison because class imbalance is common in real classification problems.

Results and Interpretation

The two models produced different results on the test set.

- Baseline DistilBERT = 66.7%
- Weighted-loss DistilBERT = 91.7%

The baseline model predicted mainly the Safe class and failed to correctly identify Risk samples.

The weighted-loss experimental model performed much better and correctly identified 3 of 4 Risk samples while maintaining perfect Safe class recall.

This suggests:

- Class imbalance was an important issue in the dataset
- Weighted loss improved minority-class detection
- Even with a small labeled dataset, performance improved through better training strategy
- More labeled data could further improve generalization

Overall, the experimental weighted-loss approach was more effective than the baseline model for this task.

Limitations and Risks

This project had several limitations:

- Only 57 labeled samples were used for modeling
- Human annotation may include subjectivity
- Binary labels simplify real rental risk complexity
- Small test set limits statistical confidence
- Listings from one city may reduce generalization to other regions

The current model should be viewed as an early prototype rather than production-ready deployment. Because of the small dataset size, model results may vary slightly across runs due to random initialization.

Ethical and Responsible Use

One important ethical concern is false positive predictions. A legitimate rental listing could be incorrectly flagged as risky, which may unfairly affect landlords or property managers.

Another concern is fairness bias. Some listings containing demographic restrictions or wording patterns may reflect broader housing discrimination issues. AI systems must be carefully reviewed to avoid reinforcing bias (Barocas et al., 2019).

The model should be used only as decision support, not as an automatic enforcement system.

Future Improvements

With more time and resources, the system could be improved by:

- Creating a much larger manually labeled dataset
- Adding multi-class labels (Safe, Suspicious, High Risk)
- Using cross-validation for stronger evaluation
- Testing larger Transformer models such as BERT or RoBERTa
- Using explainable AI tools such as SHAP or LIME
- Expanding to other Canadian cities

These steps would improve robustness and practical value.

References

Barocas, S., Hardt, M., & Narayanan, A. (2019). *Fairness and machine learning*. fairmlbook.org.

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Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. arXiv:1910.01108.